Source:

Flex Rover, Astrolab: About – Astrolab

Product Summary:

It is a multi-purpose lunar vehicle that can carry people, cargo, and science equipment. It can pick up, carry, and deposit modular payloads up to 1600 kg with more than 3 m^3 of usable payload volume. It has a robotic arm that can reach around 2 meters and is designed to manipulate payloads and perform logistical tasks such as interacting with objects on the surface, deploying equipment, or assisting with science operations. It has limbs with wheels at the ends instead of having a conventional fixed suspension. The limbs allow the rover to deal with uneven terrain and lower/raise the robot. It does have autonomous navigation capabilities, as it has navigation and hazard detection sensors. Overall this robot has a solar power system, communication equipment, can carry two astronauts, can assist with scientific and logic experiments and carry cargo from the payloader to the habitat.

Product Critique:

It can carry a lot of load, but its limit is 3 m^2 and 1000 kg and our mission is to create a robot that can carry up to 20 ft cargo modules. Additionally, this robot has not been tested in space yet it is a conceptual prototype that has been tested in California, NASA Johnson Space Center, and other lunar analog sites.

Images/Sketches:



